

Stochastic Mathematics for Finance

Week 2 Resources

Purpose of Week 2

In Week 1, we studied discrete stochastic systems such as random variables, Markov chains, and random walks. In this week, we move toward Gaussian behaviour, the Central Limit Theorem, and Monte Carlo methods.

The main conceptual goal is to understand how complicated random systems become simpler at large scales.

One of the most important ideas in probability theory is that sums of many independent random effects tend to behave like Gaussian random variables. This phenomenon is described by the Central Limit Theorem (CLT), which explains why the Normal distribution appears throughout statistics, finance, physics, and machine learning.

This week also emphasizes simulation and empirical verification. Instead of viewing probability purely theoretically, we will generate random systems computationally and observe convergence behaviour directly.

We will also introduce Monte Carlo methods, which use repeated random sampling to approximate mathematical quantities numerically. Monte Carlo simulation forms one of the computational foundations of modern quantitative finance.

By the end of this week, you should be comfortable with:

- sums of iid random variables;
- scaling behaviour of variance and standard deviation;
- the Normal distribution and Gaussian densities;
- standardization and z-scores;
- the Central Limit Theorem;
- convergence of Binomial distributions to Gaussian limits;
- simulation of Gaussian convergence in Python;
- Monte Carlo approximation methods;
- empirical verification of theoretical results.

Contents

1	Why Gaussian Behaviour Matters	4
2	Sums of Independent Random Variables	4
2.1	Interpretation	5
2.2	Variance Scaling	5
3	The Normal Distribution	5
3.1	Shape of the Gaussian Distribution	6
3.2	Why Gaussian Distributions Matter	6
4	Standard Normal Distribution	6
4.1	Standardization	6
4.2	Why Standardization Matters	6
5	The Central Limit Theorem	7
5.1	Statement of the CLT	7
5.2	What Does Convergence in Distribution Mean?	7
5.3	Why the CLT is Important	8
6	Binomial Distribution and Gaussian Convergence	8
6.1	Interpretation	8
7	Monte Carlo Methods	9
7.1	The Core Intuition	9
7.2	Why Monte Carlo Methods Work	9
7.3	A Simple Example	10
7.4	Monte Carlo Integration	10
7.5	Why Monte Carlo Integration Works	11
7.6	Monte Carlo Estimation of π	11
7.7	Monte Carlo Error Scaling	12
7.8	Why Monte Carlo Methods Are Powerful	13
7.9	Monte Carlo in Quantitative Finance	13
7.10	Limitations of Monte Carlo Methods	13
8	Simulation in Python	14
8.1	Important Libraries	14
8.2	Simulating Coin Tosses	14

8.3	Simulating Many Experiments	14
8.4	Standardized Variables	15
8.5	Plotting Standardized Histograms	15
8.6	Overlaying the Gaussian Density	15
8.7	Monte Carlo Integration	15
8.8	Monte Carlo Estimation of π	16
9	Connection to Finance	16
9.1	Volatility Scaling	16
9.2	Monte Carlo Simulation	16
10	Key Takeaways	16
11	Suggested External References	17

1. Why Gaussian Behaviour Matters

One of the most important ideas in probability theory is that complicated random systems often become simpler when viewed at large scales.

Even when individual random variables are discrete or irregular, sums of many independent random effects frequently exhibit Gaussian behaviour.

This phenomenon appears throughout:

- statistics;
- finance;
- machine learning;
- signal processing;
- physics.

The Central Limit Theorem explains why Gaussian distributions arise naturally from aggregation.

This week focuses primarily on:

- Gaussian convergence;
- variance scaling;
- empirical simulation;
- Monte Carlo methods;
- numerical approximation.

2. Sums of Independent Random Variables

Let

$$X_1, X_2, \dots, X_n$$

be iid random variables satisfying

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

Define the partial sum

$$S_n = X_1 + \dots + X_n.$$

Using linearity of expectation,

$$E[S_n] = E[X_1] + \dots + E[X_n] = n\mu.$$

If the variables are independent, covariance terms vanish, giving

$$\text{Var}(S_n) = n\sigma^2.$$

Therefore,

$$SD(S_n) = \sqrt{n}\sigma.$$

This square-root scaling is extremely important.

2.1. Interpretation

Suppose daily returns have standard deviation 1%.

Then over:

- 1 day: volatility $\approx 1\%$;
- 4 days: volatility $\approx 2\%$;
- 25 days: volatility $\approx 5\%$.

The volatility grows like $\sqrt{n}$, not linearly in n .

2.2. Variance Scaling

Suppose:

$$S_n = X_1 + \cdots + X_n.$$

Then:

$$\text{Var}(S_n) = n\sigma^2.$$

This means:

- uncertainty accumulates over time;
- fluctuations become larger at larger scales;
- randomness spreads diffusively rather than linearly.

3. The Normal Distribution

The Normal distribution, also called the Gaussian distribution, is one of the most important probability distributions in mathematics.

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The parameters are:

- μ : mean;
- σ^2 : variance;
- σ : standard deviation.

3.1. Shape of the Gaussian Distribution

The Normal distribution:

- is symmetric around the mean;
- has bell-shaped density;
- assigns high probability near the center;
- assigns lower probability to extreme values.

3.2. Why Gaussian Distributions Matter

Gaussian distributions appear because:

- sums of independent random variables converge toward Gaussians;
- many physical systems aggregate random effects;
- financial returns are often approximated using Gaussian models.

4. Standard Normal Distribution

The standard Normal distribution is

$$Z \sim \mathcal{N}(0, 1).$$

Its density is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

4.1. Standardization

If

$$X \sim \mathcal{N}(\mu, \sigma^2),$$

then

$$Z = \frac{X - \mu}{\sigma}$$

satisfies

$$Z \sim \mathcal{N}(0, 1).$$

4.2. Why Standardization Matters

Standardization allows us to:

- compare variables measured in different units;
- compute probabilities using standard Normal tables;
- analyze Gaussian convergence in the CLT.

5. The Central Limit Theorem

The Central Limit Theorem is one of the most important results in probability theory.

It states that sums of many independent random variables become approximately Gaussian after appropriate normalization.

5.1. Statement of the CLT

Let

$$X_1, X_2, \dots, X_n$$

be iid random variables satisfying

$$E[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2.$$

Define

$$S_n = X_1 + \dots + X_n.$$

Then:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}}$$

converges in distribution to

$$\mathcal{N}(0, 1).$$

Symbolically,

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow \mathcal{N}(0, 1).$$

5.2. What Does Convergence in Distribution Mean?

When we write

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow \mathcal{N}(0, 1),$$

we say that the sequence of random variables converges in distribution to the standard Normal distribution.

Intuitively, this means that as n becomes larger, the distribution of the standardized sum begins to resemble the Gaussian distribution more and more closely.

Importantly, convergence in distribution does **not** mean that the random variables become equal. Instead, it means that their probability distributions become increasingly similar.

For example, suppose:

$$X_i \sim \text{Bernoulli}(1/2).$$

Then each individual variable only takes values 0 or 1, which is very far from Gaussian. However, after summing many independent copies and standardizing, the resulting distribution begins to look bell-shaped.

This can be observed empirically:

- for small n , histograms look discrete and irregular;
- for moderate n , they begin smoothing out;

- for large n , they closely resemble the Gaussian density.

Convergence in distribution is therefore about the *shape* of the distribution becoming Gaussian in the large- n limit.

5.3. Why the CLT is Important

The CLT explains why Gaussian distributions appear in:

- statistics;
- finance;
- physics;
- machine learning;
- signal processing.

Even if the underlying variables are not Gaussian, their aggregated behaviour often becomes approximately Gaussian.

This is one of the deepest universality principles in probability theory.

6. Binomial Distribution and Gaussian Convergence

Suppose

$$X_n \sim \text{Binomial}(n, p).$$

Then:

$$E[X_n] = np,$$

and

$$\text{Var}(X_n) = np(1 - p).$$

The standardized variable

$$\frac{X_n - np}{\sqrt{np(1 - p)}}$$

approaches

$$\mathcal{N}(0, 1)$$

as $n \rightarrow \infty$.

6.1. Interpretation

A Binomial random variable counts the number of successes in many independent Bernoulli trials.

As the number of trials grows:

- the discrete distribution becomes smoother;
- the histogram becomes bell-shaped;
- Gaussian behaviour emerges.

7. Monte Carlo Methods

Monte Carlo methods are computational techniques that use randomness and repeated simulation to approximate mathematical quantities.

The fundamental idea is surprisingly simple:

If a quantity can be interpreted probabilistically, we can estimate it using random samples.

Instead of solving a problem analytically, we:

1. generate many random outcomes;
2. compute the quantity of interest for each outcome;
3. average the results.

Monte Carlo methods are among the most important computational tools in:

- quantitative finance;
- machine learning;
- computational physics;
- Bayesian statistics;
- engineering;
- scientific computing.

7.1. The Core Intuition

Suppose we want to estimate the average height of students in a university.

Measuring every student individually may be difficult.

Instead, we:

- randomly sample a subset of students;
- compute the average height of the sample;
- use this as an estimate of the true average.

As the sample size increases, the sample average becomes increasingly accurate.

Monte Carlo methods apply this exact same philosophy to mathematical and probabilistic problems.

7.2. Why Monte Carlo Methods Work

Monte Carlo methods work because of the Law of Large Numbers.

Suppose:

$$X_1, X_2, \dots, X_N$$

are iid random variables satisfying:

$$E[X_i] = \mu.$$

Then:

$$\frac{1}{N} \sum_{i=1}^N X_i \rightarrow \mu$$

as:

$$N \rightarrow \infty.$$

This means that averages of many independent samples stabilize around the true expected value.

Although individual random samples may fluctuate significantly, averaging many samples reduces randomness and reveals the underlying statistical structure.

This is the mathematical foundation of Monte Carlo simulation.

7.3. A Simple Example

Suppose we repeatedly toss a fair coin.

Define:

$$X_i = \begin{cases} 1 & \text{Heads} \\ 0 & \text{Tails} \end{cases}$$

The true probability of heads is:

$$E[X_i] = 0.5.$$

Now suppose we perform many simulations and compute:

$$\frac{1}{N} \sum_{i=1}^N X_i.$$

This quantity represents the fraction of heads observed experimentally.

For small N , the estimate may fluctuate significantly.

However, as N becomes larger:

$$\frac{1}{N} \sum_{i=1}^N X_i \approx 0.5.$$

Monte Carlo methods use this same averaging principle in far more complicated settings.

7.4. Monte Carlo Integration

One of the most important applications of Monte Carlo methods is numerical integration.

Suppose we want to approximate:

$$I = \int_0^1 f(x) dx.$$

Analytically computing the integral may be difficult or impossible.

Instead, we generate random variables:

$$U_1, U_2, \dots, U_N \sim \text{Uniform}(0, 1).$$

We then compute:

$$\frac{1}{N} \sum_{i=1}^N f(U_i).$$

This quantity approximates:

$$\int_0^1 f(x) dx.$$

Thus:

$$I \approx \frac{1}{N} \sum_{i=1}^N f(U_i).$$

Instead of solving the integral directly, we estimate it statistically.

7.5. Why Monte Carlo Integration Works

If:

$$U \sim \text{Uniform}(0, 1),$$

then:

$$E[f(U)] = \int_0^1 f(x) dx.$$

Therefore:

$$\frac{1}{N} \sum_{i=1}^N f(U_i)$$

is simply a sample average estimating:

$$E[f(U)].$$

By the Law of Large Numbers:

$$\frac{1}{N} \sum_{i=1}^N f(U_i) \rightarrow \int_0^1 f(x) dx.$$

This explains mathematically why Monte Carlo integration converges toward the true integral.

7.6. Monte Carlo Estimation of π

A classic Monte Carlo example is estimating π geometrically.

Consider the square:

$$[-1, 1] \times [-1, 1].$$

Inside the square lies the unit circle:

$$x^2 + y^2 \leq 1.$$

The area of the square is:

$$4.$$

The area of the unit circle is:

$$\pi.$$

Therefore:

$$\frac{\text{Area of Circle}}{\text{Area of Square}} = \frac{\pi}{4}.$$

Now generate random points uniformly inside the square.

The fraction of points landing inside the circle approximates:

$$\frac{\pi}{4}.$$

Thus:

$$\pi \approx 4 \times (\text{fraction of points inside circle}).$$

This is a beautiful example of randomness being used to approximate a deterministic quantity.

7.7. Monte Carlo Error Scaling

Monte Carlo estimators improve gradually as the number of samples increases.

The typical error behaves like:

$$\frac{1}{\sqrt{N}}.$$

This scaling is extremely important.

For example:

- improving accuracy by a factor of 10 requires roughly 100 times more samples;
- reducing noise significantly may require very large simulations;
- computational efficiency becomes extremely important.

This motivates the development of:

- variance reduction techniques;
- importance sampling;
- antithetic variates;
- quasi-Monte Carlo methods.

7.8. Why Monte Carlo Methods Are Powerful

Monte Carlo methods are especially powerful because they remain effective even when problems become extremely complicated.

Traditional numerical methods often struggle in:

- high-dimensional spaces;
- stochastic systems;
- complicated geometries;
- nonlinear models.

Monte Carlo methods often remain feasible in these settings.

This makes them one of the most versatile computational tools in modern mathematics.

7.9. Monte Carlo in Quantitative Finance

Modern quantitative finance relies heavily on Monte Carlo simulation.

Applications include:

- option pricing;
- portfolio simulation;
- Value-at-Risk estimation;
- derivative pricing;
- stress testing;
- risk management.

For many realistic financial models:

- analytical formulas do not exist;
- market dynamics are stochastic;
- exact computation becomes impossible.

In these situations, Monte Carlo simulation becomes one of the only practical computational approaches.

7.10. Limitations of Monte Carlo Methods

Although Monte Carlo methods are extremely useful, they also have limitations.

- convergence may be slow;
- large simulations can be computationally expensive;

- estimates contain statistical noise;
- poor random number generation can introduce bias.

Nevertheless, Monte Carlo methods remain one of the foundational tools of computational finance and scientific computing.

8. Simulation in Python

Simulation is the core computational focus of this week.

The goal is not only to understand the Central Limit Theorem theoretically, but also to observe Gaussian convergence empirically.

8.1. Important Libraries

- `numpy`
- `matplotlib`
- `scipy.stats`

8.2. Simulating Coin Tosses

```
1 import numpy as np
2
3 n = 1000
4
5 X = np.random.choice([0, 1], size=n)
6
7 print(X[:20])
```

8.3. Simulating Many Experiments

```
1 import numpy as np
2
3 N = 10000
4 n = 50
5
6 samples = np.random.binomial(
7     1,
8     0.5,
9     size=(N, n)
10 )
11
12 S = samples.sum(axis=1)
13
14 print(S[:10])
```

8.4. Standardized Variables

```
1 Z = (S - n * 0.5) / np.sqrt(n * 0.25)
```

8.5. Plotting Standardized Histograms

```
1 import matplotlib.pyplot as plt
2
3 plt.hist(
4     Z,
5     bins=50,
6     density=True
7 )
8
9 plt.show()
```

8.6. Overlaying the Gaussian Density

```
1 from scipy.stats import norm
2
3 x = np.linspace(-4, 4, 1000)
4
5 plt.hist(
6     Z,
7     bins=50,
8     density=True
9 )
10
11 plt.plot(
12     x,
13     norm.pdf(x)
14 )
15
16 plt.show()
```

8.7. Monte Carlo Integration

```
1 import numpy as np
2
3 N = 100000
4
5 U = np.random.uniform(
6     0,
7     1,
8     size=N
9 )
10
11 estimate = np.mean(U**2)
12
13 print(estimate)
```

This approximates:

$$\int_0^1 x^2 dx = \frac{1}{3}.$$

8.8. Monte Carlo Estimation of π

```
1 import numpy as np
2
3 N = 100000
4
5 x = np.random.uniform(-1, 1, size=N)
6 y = np.random.uniform(-1, 1, size=N)
7
8 inside = (x**2 + y**2) <= 1
9
10 pi_estimate = 4 * np.mean(inside)
11
12 print(pi_estimate)
```

9. Connection to Finance

The ideas in this week appear throughout quantitative finance.

9.1. Volatility Scaling

If daily returns have variance σ^2 , then:

$$\text{Var}(\text{monthly return}) \approx n\sigma^2.$$

Thus:

$$SD(\text{monthly return}) \approx \sqrt{n}\sigma.$$

9.2. Monte Carlo Simulation

Modern quantitative finance relies extensively on Monte Carlo simulation for:

- option pricing;
- risk estimation;
- portfolio simulation;
- scenario generation.

10. Key Takeaways

- Sums of iid random variables exhibit $\sqrt{n}$ scaling.
- Variance grows linearly with time.

- Gaussian distributions naturally emerge from aggregation.
- The Central Limit Theorem explains universal Gaussian behaviour.
- Binomial distributions converge toward Gaussian distributions.
- Monte Carlo methods use randomness for numerical approximation.
- Simulation provides empirical verification of theoretical probability results.

11. Suggested External References

- Sheldon Ross — *Introduction to Probability Models*
- Rick Durrett — *Probability: Theory and Examples*
- Glasserman — *Monte Carlo Methods in Financial Engineering*
- MIT OpenCourseWare Probability Lectures
- 3Blue1Brown videos on the Central Limit Theorem

Week 2 Summary

This week introduced one of the deepest ideas in probability theory: large collections of independent random effects become approximately Gaussian.

We studied:

- Gaussian distributions;
- the Central Limit Theorem;
- variance scaling;
- Monte Carlo methods;
- empirical verification through simulation.

These ideas form the mathematical bridge between elementary probability and modern stochastic modelling.